Probabilistic Machine Learning

Luca Bortolussi

Department of Mathematics, Informatics, and Geosciences
University of Trieste

Office 3.33, third floor, Via Economo
lbortolussi@units.it

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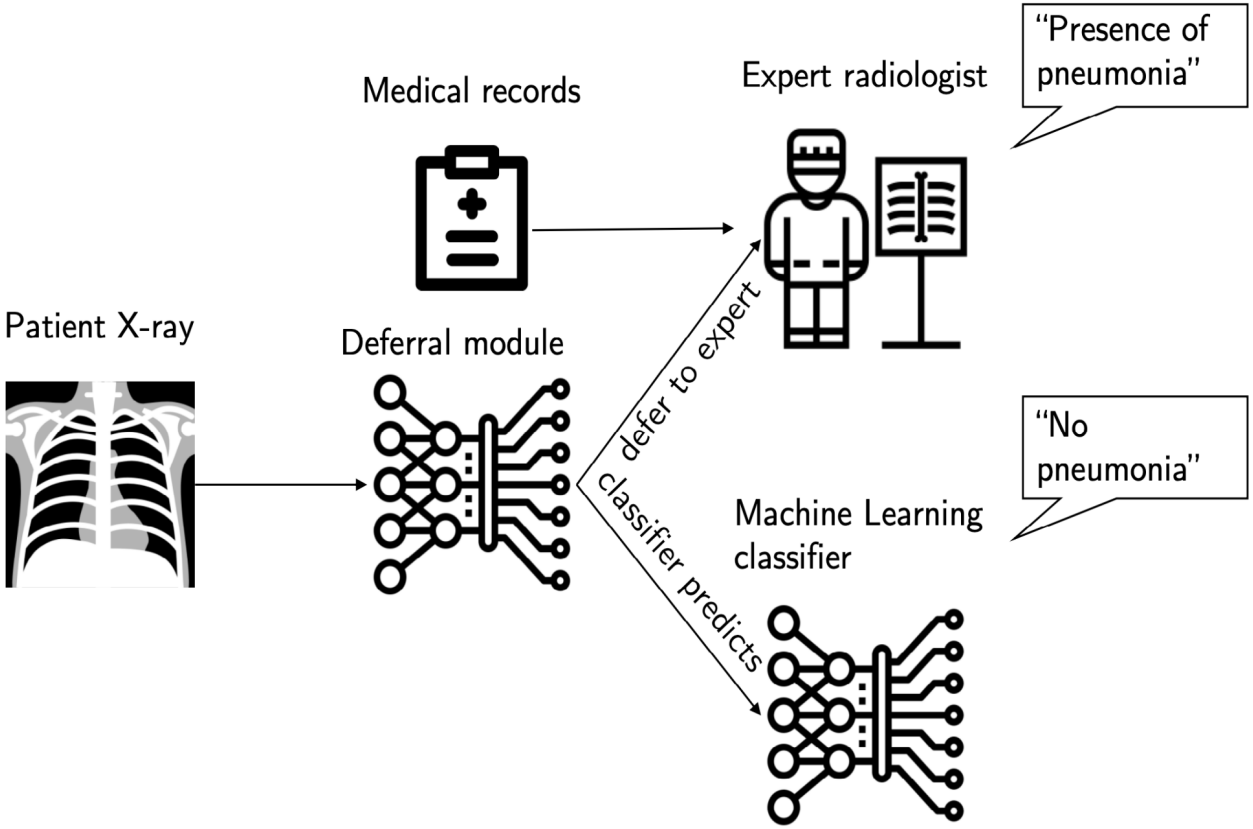
Data Science and Artificial Intelligence

An application scenario



Machine Learning

What is Machine Learning all about?

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Learning Models

Machine Learning

What is a Model?

Discuss among you 5 minutes and then answer...

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- All modelling usually starts by defining a *family* of models indexed by some parameters, which are tweaked to reflect how well the feature of interest is captured.
- Machine learning deals with algorithms for automatic selection of a model from observations of the system.

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- Epistemic uncertainty: describes lack of knowledge about a phenomenon.

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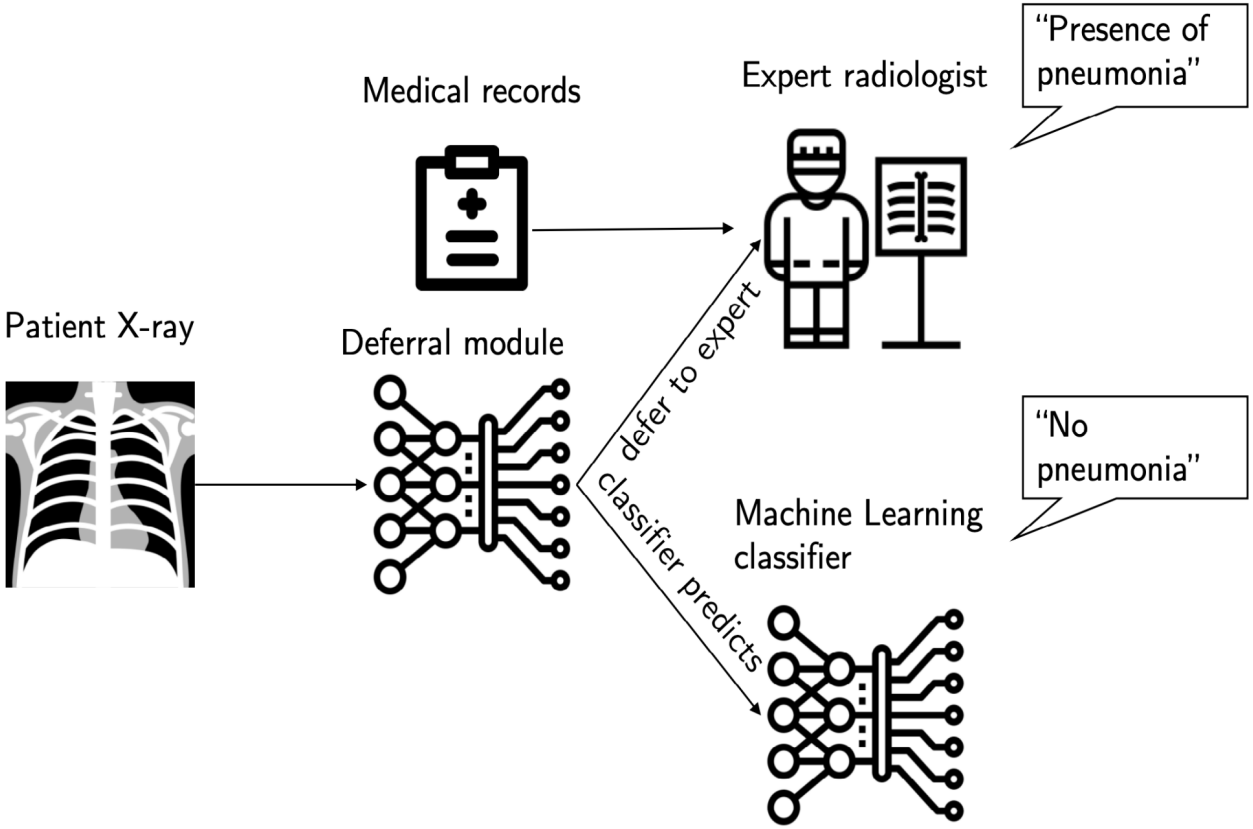

A rough classification

Machine learning explores the study and construction of algorithms that can learn from and make predictions on data. [source: wikipedia]

- Supervised learning: learn a model from input-output data. The goal is to predict a the (most-likely) output value for a new, unobserved, input. We distinguish
  - Regression (continuous output)
  - Classification (binary/ discrete output)
- Unsupervised learning: extract information/ learn a model from input-only data
- Reinforcement Learning: find suitable actions to take in a given uncertain situation in order to maximize a reward.

Generative and Discriminative models

- **Generative models** aim at describing the full probability distribution of inputs x or input/output pairs x, y:

$$p(x, y) = p(x)p(y \mid x)$$

- **Discriminative learning** aims at describing the conditional probability of output given the input, or a statistics/ function of such probability:

$$p(y \mid x) \text{ or } y = f(x)$$

Machine learning

- Supervised learning: learn $p(y \mid x)$ or the best prediction (discriminant) $y = f(x)$
- Unsupervised learning: learn $p(x)$ or some properties of it (e.g. clusters)
- Data generation: learn a model of $p(x)$ and sample new elements from it (or from $p(x \mid y)$).

Inference and Estimation

Two central concepts for probabilistic machine learning

- Inference: compute marginals and conditional probability distributions applying the laws of probability.
- Estimation: Given data and a family of models, find the best parameters/models for the data.

In the Bayesian world: estimation $\approx$ inference.

Course organization

- Class lectures: Class lectures will be recorded. Students are in any case strongly recommended to attend the course in presence, and ask questions during lectures.
- Online lectures: For students that cannot attend the course in presence, recorded lectures are available online. For some parts of the course, there are more structured lectures.

Course topics - I

- (Basics of probability and probabilistic inference)
- Probabilistic formulation of learning (Empirical Risk Minimization and PAC learning).
- Graphical models
- Inference with graphical models: belief propagation
- Hidden Markov Models for sequential data
- Bayesian Linear Regression and Classification, Laplace approximation, Model Selection;
- Kernel Regression and Kernel functions, Gaussian Processes for Regression (hints)

Course topics - II

- Monte Carlo sampling
- Expectation Maximisation and Variational Inference.
- Bayesian Neural Networks
- Generative Modelling: Variationa Autoencoders and Diffusion Processes.

Lab

Lab classes are fundamental to learn how to apply the notions discussed in the course, working out some challenging scenario. We will learn how to use Pyro (a torch based probabilistic programming language).

Always bring your own laptop...

Lab will be learn by doing, with a lot of self learning. Working in groups is ok. You can also work on your own data and problems (from Kaggle, from your past courses).

Exam

- Final team project 2-3 people, with presentation: presentation, implementation and testing of a theoretical methodology, or analysis of a dataset, using methods seen during the course or some powered up version (hence: no standard deep learning).
- Teams will be formed by you but under some diversity constraints.
- Topics need to be proposed by the group and accepted by the professor.
- An individual oral colloquium with questions on stuff seen during the course, including theory.

Coordinates

Teams

All material will be in a dedicated Team of the course (not on course catalogue). Register with this code: **s66ed6a**.

Where can you find me?

- Room 3.33, E building (Via Economo), 3rd floor - email me first at `lbortolussi@units.it`.

Questions?